

- (a) The table shows the time, (in minutes), that each of 120 students from YT Secondary School spent on social media daily.

Time (t minutes)	$30 < t \leq 60$	$60 < t \leq 90$	$90 < t \leq 120$	$120 < t \leq 150$	$150 < t \leq 180$
Number of students	5	25	33	47	10

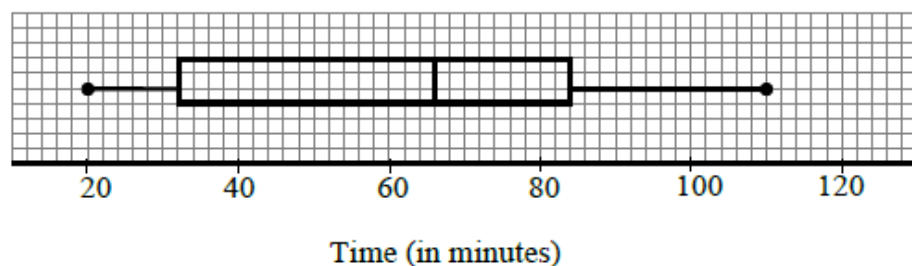
- (i) Which interval contains the median time spent on social media?

[1]

Calculate an estimate of

- (ii) the mean amount of time spent on social media daily, [1]
 (iii) the standard deviation of the time spent on social media daily. [1]
 (iv) 2 students from YT Secondary school are chosen at random. Find the probability that one of them spent more than 2.5 hours while the other spent less than 1 hour on social media everyday. [2]

- (b) The box-and-whisker plot below shows the time, (in minutes), that each of 200 students from AB Secondary School spent on social media daily.

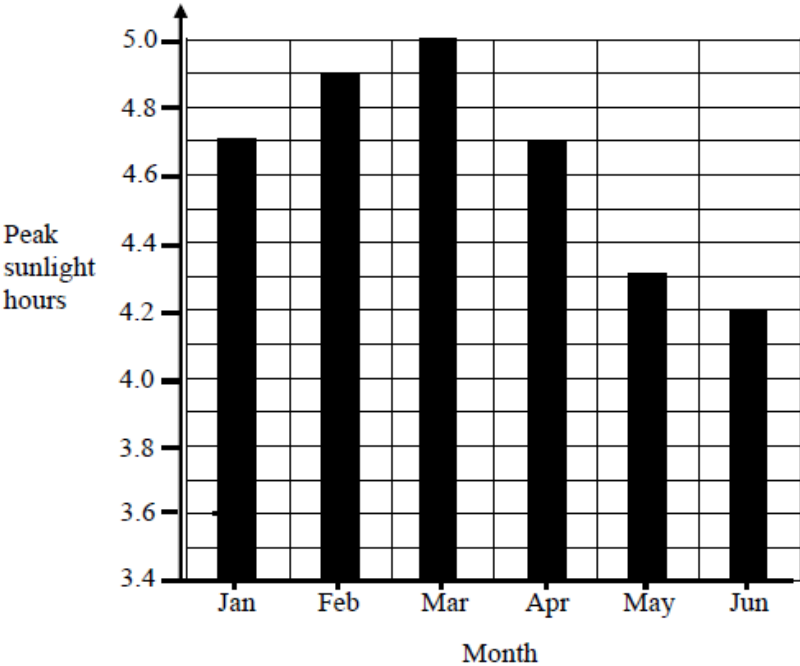


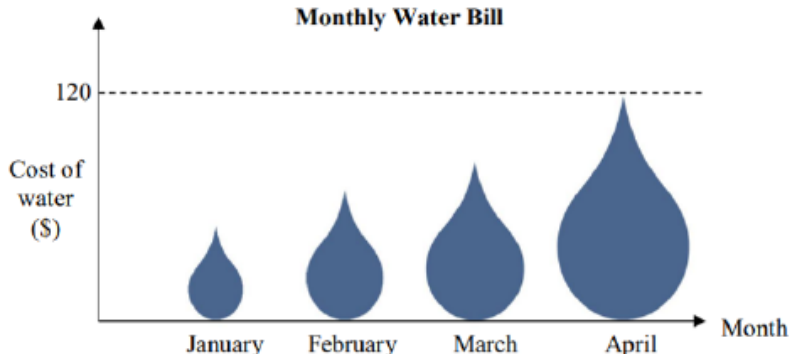
- (i) $\frac{3}{4}$ of the students spent more than k minutes on social media daily. Find the value of k . [1]
 (ii) A student from AB Secondary school is chosen at random. Find the probability that he spent more than 84 minutes on social media daily. [1]
 (iii) 200 students from PQR Secondary School were also asked the amount of time (in minutes) they spent on social media daily. The results are summarized in the table.

Median	72
Interquartile Range	65

Make a comment comparing the averages and a comment comparing the distributions of the times, (in minutes), spent by the students in AB and PQR Secondary schools respectively.
 Use figures to support your answers.

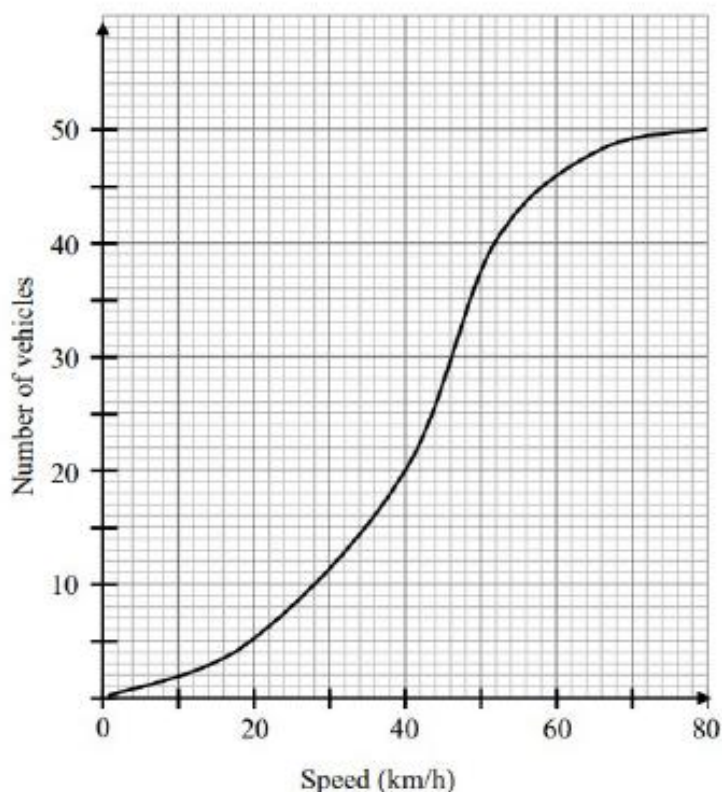
[2]

2	SCGS/2025/I/09 The graph shows the average daily peak sunlight hours in Singapore over six months.  <table border="1"> <caption>Peak Sunlight Hours Data</caption> <thead> <tr> <th>Month</th> <th>Peak sunlight hours</th> </tr> </thead> <tbody> <tr> <td>Jan</td> <td>4.7</td> </tr> <tr> <td>Feb</td> <td>4.9</td> </tr> <tr> <td>Mar</td> <td>5.0</td> </tr> <tr> <td>Apr</td> <td>4.7</td> </tr> <tr> <td>May</td> <td>4.3</td> </tr> <tr> <td>Jun</td> <td>4.2</td> </tr> </tbody> </table> Mary said that the average daily peak sunlight hours in March is twice as long as that in June. Explain why Mary may have been misled to give this wrong conclusion.	Month	Peak sunlight hours	Jan	4.7	Feb	4.9	Mar	5.0	Apr	4.7	May	4.3	Jun	4.2
Month	Peak sunlight hours														
Jan	4.7														
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	[2]														
3	SCGS/2025/I/19 The stem-and-leaf diagram shows the masses, in kilograms, of a group of 21 students. <pre> 4 6 6 6 7 9 5 0 0 1 1 1 2 5 5 6 6 6 6 7 9 6 2 8 x </pre> Key : 4 6 means 46 kg for the students (a) The range of the masses for the students is 36 kg. Find the value of x. (b) Explain why the median is a better measure of central tendency of masses for the students. (c) Find the median mass for the students. (d) Find the interquartile range of the masses for the students.														
	[1] [1] [1] [2]														

4	SCGS/2025/II/03																
	The time, h hours, spent working at home by each of 110 adults in a week is recorded. The table shows information about these times. <table><tr><th>Time (h hours)</th><th>$0 < h \leq 10$</th><th>$10 < h \leq 20$</th><th>$20 < h \leq 30$</th><th>$30 < h \leq 40$</th><th>$40 < h \leq 50$</th></tr><tr><th>Frequency</th><td>13</td><td>45</td><td>24</td><td>12</td><td>16</td></tr></table> (a) (i) Write down the interval containing the median time. [1] (ii) Calculate an estimate of the mean time. [1] (iii) Calculate an estimate of the standard deviation of the times. [1] (b) (i) One of these 110 adults is chosen at random. Find, as a fraction in its lowest terms, the probability that this adult spent more than 30 hours working at home. [1] (ii) Two adults are chosen at random from those who spent 30 hours or less working at home. Calculate the probability that at least one of them spent more than 10 hours working at home. [2] (c) The time spent working at the office by the same group of adults in the same week is also recorded. The results are summarised in the table. <table><tr><td>Mean</td><td>24 hours</td></tr><tr><td>Standard Deviation</td><td>8.5 hours</td></tr></table> Make two comments comparing the times taken by the adults to work at home and at the office. [2]	Time (h hours)	$0 < h \leq 10$	$10 < h \leq 20$	$20 < h \leq 30$	$30 < h \leq 40$	$40 < h \leq 50$	Frequency	13	45	24	12	16	Mean	24 hours	Standard Deviation	8.5 hours
Time (h hours)	$0 < h \leq 10$	$10 < h \leq 20$	$20 < h \leq 30$	$30 < h \leq 40$	$40 < h \leq 50$												
Frequency	13	45	24	12	16												
Mean	24 hours																
Standard Deviation	8.5 hours																
5	CCHY/2025/I/11																
	Mrs Lim draws this graph to show her monthly water bill for the first four months of the year 2024. Monthly Water Bill  State one aspect of the graph that may be misleading and explain how this may lead to a misinterpretation of the graph. [2]																

6 CCHY/2025/I/17

The cumulative frequency graph shows the speed of 50 vehicles passing by a narrow road within 5 minutes.



- (a) If 8% of the 50 vehicles received a speeding ticket, find the speed limit on this road. [1]
- (b) Find the interquartile range of the speed of the vehicles. [2]
- (c) It was said that more than 50% of the vehicles are travelling at least 45 km/h when passing through the narrow road. Explain, with reason, whether you agree with the statement. [1]

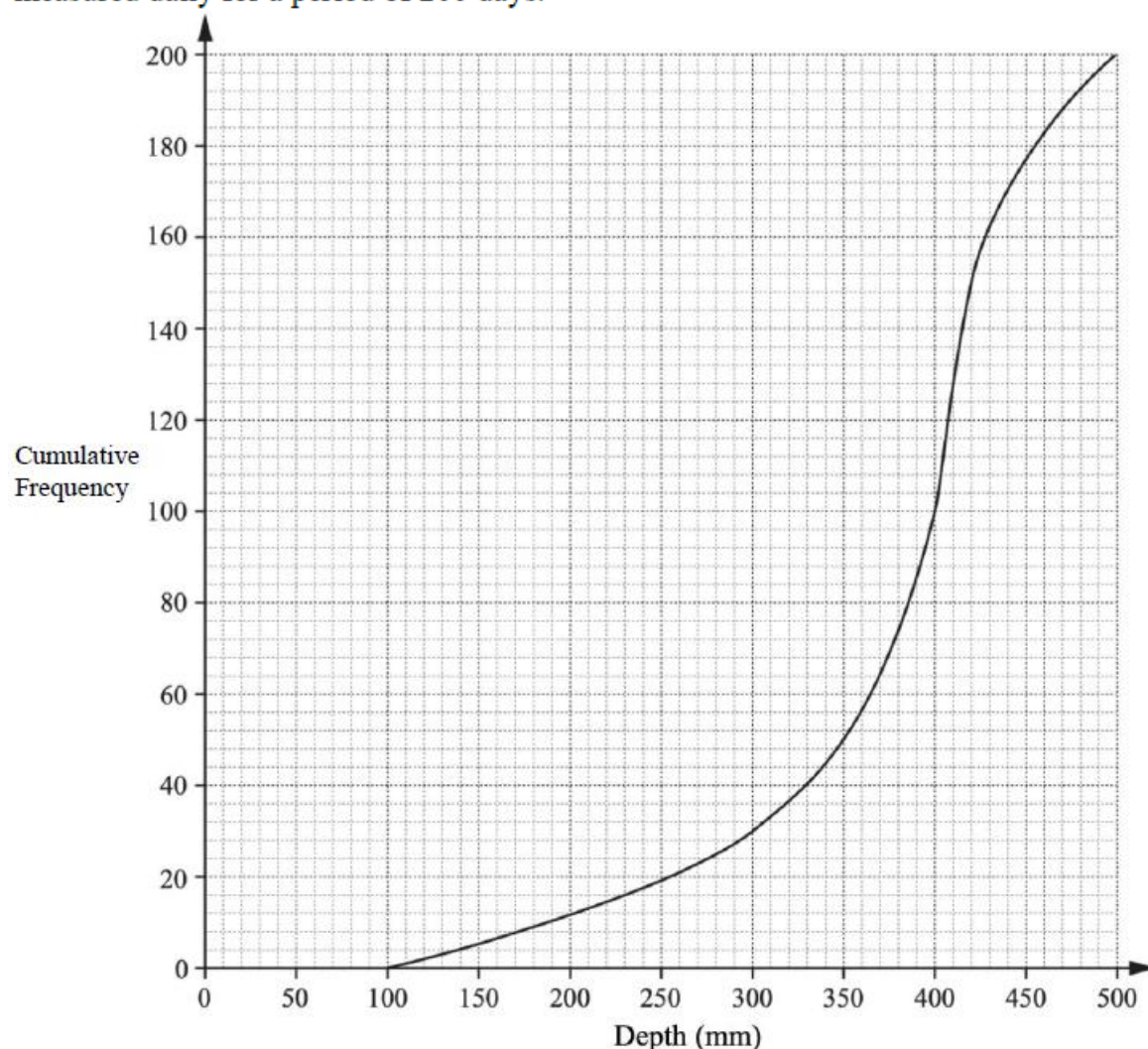
7 CCHY/2025/I/21

The table shows the quiz scores for a group of 30 students.

Quiz score, m	Frequency
$0 < m \leq 5$	2
$5 < m \leq 10$	5
$10 < m \leq 15$	15
$15 < m \leq 20$	8

- (a) Calculate an estimate for
- (i) the mean scores of the students. [1]
- (ii) the standard deviation of the quiz scores. [1]
- (b) A student who was sick did a make-up for the quiz.
The new mean mark was 13.
Given that the original median was 12, what can be deduced about the new median? [1]

The cumulative frequency curve shows the distribution of the depth, d mm, of a canal measured daily for a period of 200 days.



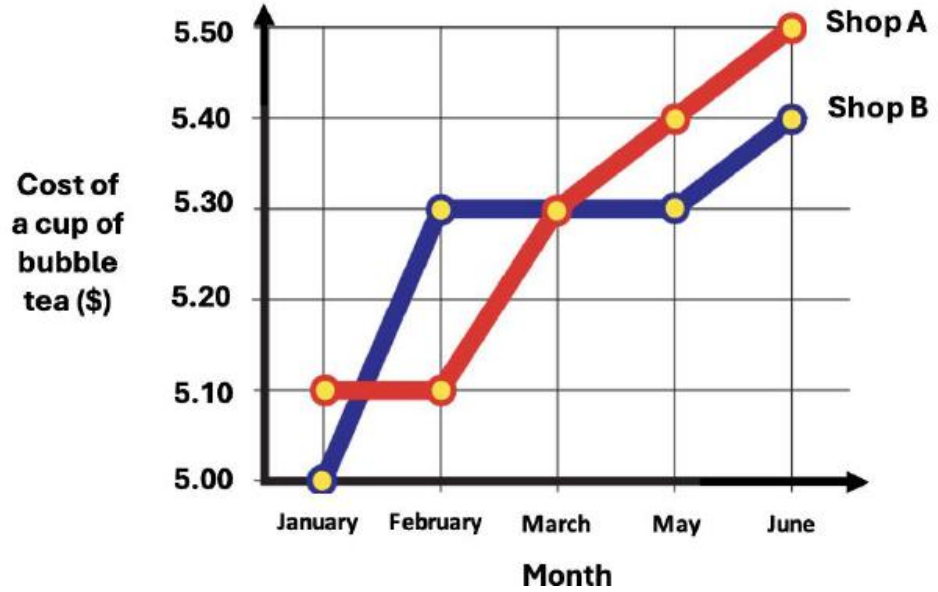
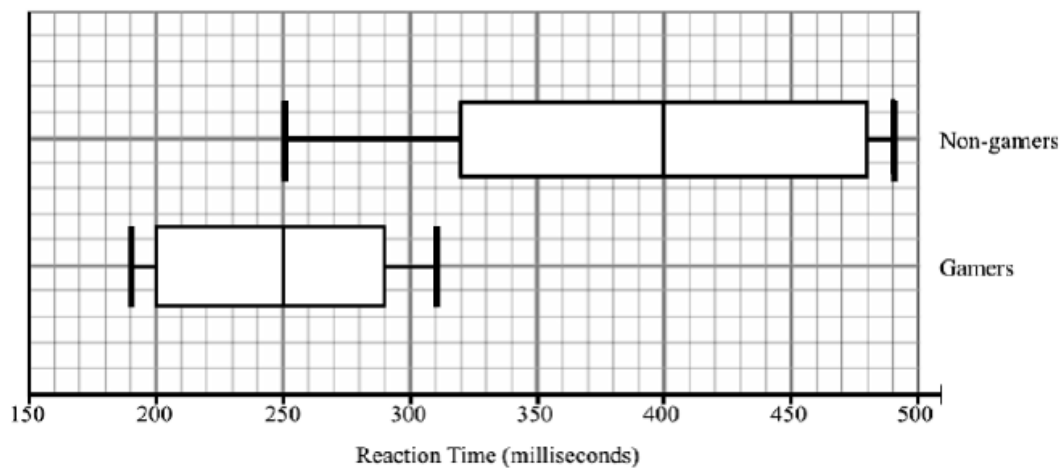
(a) Use the curve to estimate

- (i) the median depth of the canal, [1]
- (ii) the interquartile range of the depth, [1]
- (iii) the depth at the 30th percentile, [1]
- (iv) the number of days when the depth of the canal was at least 450 mm. [1]

(b) The information on the cumulative frequency curve is shown in this table.

Depth (mm)	$100 < d \leq 200$	$200 < d \leq 300$	$300 < d \leq 400$	$400 < d \leq 500$
No. of days	12	m	n	100

- (i) Find the values of m and n . [1]
- (ii) Calculate an estimate of the mean depth of the canal. [1]

9	SST/2025/I/21
	Cost of Bubble Tea has Risen Drastically  State one aspect of the graph that may be misleading and explain how this may lead to a misinterpretation of the graph.
	[2]
10	SST/2025/II/7(a)
	(a) A study records the reaction times of gamers and non-gamers. The box-and-whisker plots show the distributions of the results.  (i) There are 15 non-gamers who had a reaction time slower than 480 milliseconds. Work out the number of non-gamers in the study. (ii) Make a comment comparing the averages and a comment comparing the distributions of the times taken by the gamers and non-gamers. Use figures to support your answers.
	[1]
	[2]

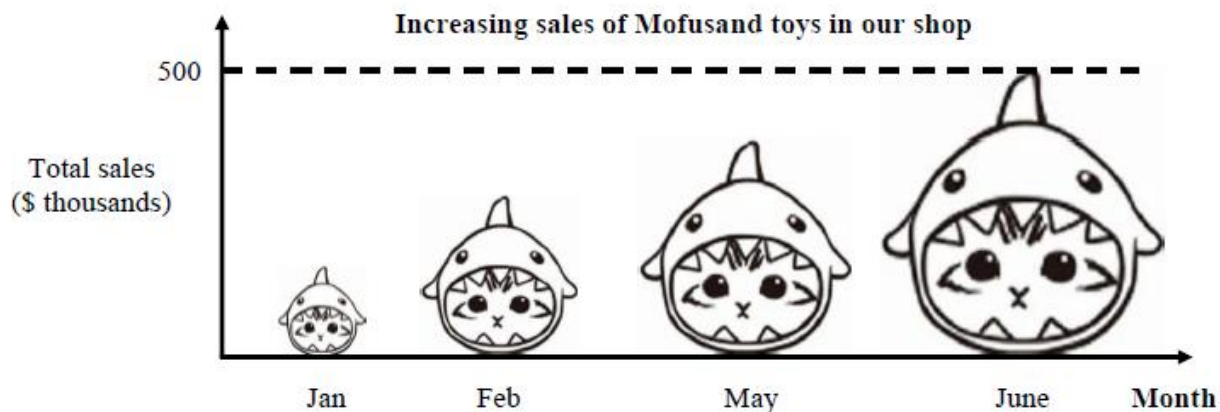
- (a) Ten boys and nine girls took Mathematics test.
The marks are shown in the stem-and-leaf diagram below.

Boys		Girls
2	1	9
5 4 3 2	2	0
5 1	3	4 5 5 5
y 7 4	4	1 6 8

Key (Boys): 2|1 means 12

Key (Girls): 1|9 means 19

- (i) State the modal mark for the girls. [1]
 (ii) The range of marks for the boys is 37. Find y . [1]
 (iii) Find the interquartile of the marks for the girls. [2]
 (b) A toy shop worker came up with the sales chart below.



State one aspect of the graph that may be misleading and explain how this may lead to a misinterpretation of the graph.

[2]

12	ACSBR/2025/I/17 The table shows the monthly salaries of a group of 20 employees. <table border="1"> <thead> <tr> <th>Monthly salary, m (\$)</th><th>Frequency</th></tr> </thead> <tbody> <tr> <td>$3000 \leq m < 3200$</td><td>3</td></tr> <tr> <td>$3200 \leq m < 3400$</td><td>7</td></tr> <tr> <td>$3400 \leq m < 3600$</td><td>8</td></tr> <tr> <td>$3600 \leq m < 3800$</td><td>2</td></tr> </tbody> </table> (a) Calculate an estimate for (i) the mean monthly salary of the employees, [1] (ii) the standard deviation of the monthly salaries. [1] (b) Each of the employees is to be given a salary increase of \$100 per month. Explain how the mean and standard deviation will change after this increase. [1]	Monthly salary, m (\$)	Frequency	$3000 \leq m < 3200$	3	$3200 \leq m < 3400$	7	$3400 \leq m < 3600$	8	$3600 \leq m < 3800$	2
Monthly salary, m (\$)	Frequency										
$3000 \leq m < 3200$	3										
$3200 \leq m < 3400$	7										
$3400 \leq m < 3600$	8										
$3600 \leq m < 3800$	2										
13	ACSBR/2025/II/19 For a school project, Akif drew this graph to show his annual electricity bill for each of the last 3 years. State one aspect of the graph that may be misleading and explain how this may lead to a misinterpretation of the graph. [2]										

Solution

1	(ai) $90 < t \leq 120$ (aii) Mean = 113 [B1] (aiii) Standard Deviation = 30.4 [B1] (aiv) $\frac{5}{120} \times \frac{10}{119} \times 2 = \frac{5}{714}$ [M1] [A1] for x2 and answer (bi) $k = 32$ [B1] (bii) $\frac{1}{4}$ [B1] (biii) Since $66 < 72$, median time spent on social media by students from XYZ secondary school is higher than that from AB secondary school. Students from XYZ secondary school spent more time on social media than students from AB secondary school. Since $52 < 65$, interquartile range of time spent on social media by AB secondary school is smaller than that from XYZ secondary school. The time spent on social media by students from AB secondary school is more consistent than students from XYZ secondary school.
2	Mary was comparing the heights of the bars where March was twice as tall as June. This is incorrect as the axis for peak sunlight hours (or vertical axis) did not start from zero.
3	(a) Highest mass = $46 + 36 = 82$ kg $\therefore x = 2$ (b) There is an extreme value of 82 kg which will cause the mean to be higher than expected. (c) 52 kg (d) IQR = $56 - 49.5$ $= 6.5$ kg

4	ai $10 < h \leq 20$ aii Mean = 22.5 hours (3 s.f.) aiii Standard deviation = 12.3 hours (3) bi $P(\text{one} > 30 \text{ hours}) = \frac{28}{110} = \frac{14}{55}$ bii <u>Method 1</u> $P(\text{at least one} > 10 \text{ hours})$ $= 1 - P(\text{both} \leq 10 \text{ hours})$ $= 1 - \left(\frac{13}{82}\right)\left(\frac{12}{81}\right)$ $= \frac{1081}{1107}$ bii <u>Method 2</u> $P(\text{at least one} > 10 \text{ hours})$ $= P(> 10, \leq 10) \times 2 + P(> 10, > 10)$ $= \left(\frac{69}{82}\right)\left(\frac{13}{81}\right) \times 2 + \left(\frac{69}{82}\right)\left(\frac{68}{81}\right)$ $= \frac{1081}{1107}$ c Since the mean time of 22.5 hours spent working at home is less than the 24 hours spent working at the office, the adults generally spent more time working at the office than at home. Since the standard deviation of 12.3 hours working at home is greater than the 8.5 hours working at the office, the times spent working at home is more spread out and less consistent.
5	No equal intervals indicated on the vertical axis and if the values start from 0. Unable to read the cost of water in the months before April. OR Not clear if the size of the water droplets, is referring to the area of the droplets or the height of the droplets. Unable to determine the annual water bill.
6	(a) 60 km/h (b) $50 - 32$ or $50 - 31$ $= 18$ $= 19$ (c) Disagree. 50% of the vehicles are travelling at a speed less than 44km/h.

7	(ai) $\text{mean} = \frac{2.5 \times 2 + 7.5 \times 5 + 12.5 \times 15 + 17.5 \times 8}{30}$ $= 12\frac{1}{3}$ (aii) $\text{s.d.} = \sqrt{\frac{2.5^2 \times 2 + 7.5^2 \times 5 + 12.5^2 \times 15 + 17.5^2 \times 8}{30} - \left(12\frac{1}{3}\right)^2}$ $= 4.18$ (b) <i>Answer</i> The new median can be either 12 or larger than 12. OR the new median cannot be lower than 12.
8	(ai) Median = 400 mm. (aii) $= 420 - 350$ $= 70 \text{ mm}$ (aiii) Reading at 60 days = 365 mm (aiv) Number of days when the depth of the canal was at least 450 mm $= 200 - 178$ $= 22$ (bi) $m = 30 - 12 = 18$ $n = 70$ (bii) Estimated mean $= \frac{(150 \times 12) + (250 \times 18) + (350 \times 70) + (450 \times 100)}{200}$ $= \frac{1800 + 4500 + 24500 + 45000}{200}$ $= \frac{77000}{200}$ $= 379 \text{ mm}$

9	Possible answers: (i) The vertical axis did not start from zero. This exaggerates the differences between the price difference of \$0.10. (ii) The title is biased. This misguides the readers to think that the price of bubble tea has increased significantly and does not allow readers to make their own judgement. (iii) Information from the month of April is missing. The reader may assume that the price of bubble tea for Shop B has remained constant compared to the the price of bubble tea for Shop A which has increased.	
10	(ai) $15 \times 4/3 = 20$ (aii) Gamers have a lower <i>median</i> of <u>250</u> milliseconds compared to <u>400</u> milliseconds for non-gamers. Hence, gamers generally had <u>faster</u> reaction times. The <i>interquartile range</i> for gamers of $290 - 200 = 90$ milliseconds is smaller than that for non-gamers of $480 - 320 = 160$ milliseconds. Hence, non-gamers had a <u>wider spread</u> in reaction times, showing <u>less consistency</u>.	
11	(ai) 35 (aii) 9 (aiii) 15.5 (b) Misleading aspect: Missing sales figures for March and April (horizontal axis) Missing numbers on vertical axis Misinterpretation: Incomplete data to support "Increasing sales..." Reader might use area of picture to compare the sales figure Area of each picture is not directly proportional to the height	
12	(ai) 3390 (aii) 173 (b) Mean will increase by \$100 but no change in standard deviation.	
13	The y axis of the graph does not start from zero. Comparing the electricity bill from 2022 to 2023, readers might misinterpret the graph by thinking that electricity bill looks like it has doubled when in fact it is a 33.3% increase.	